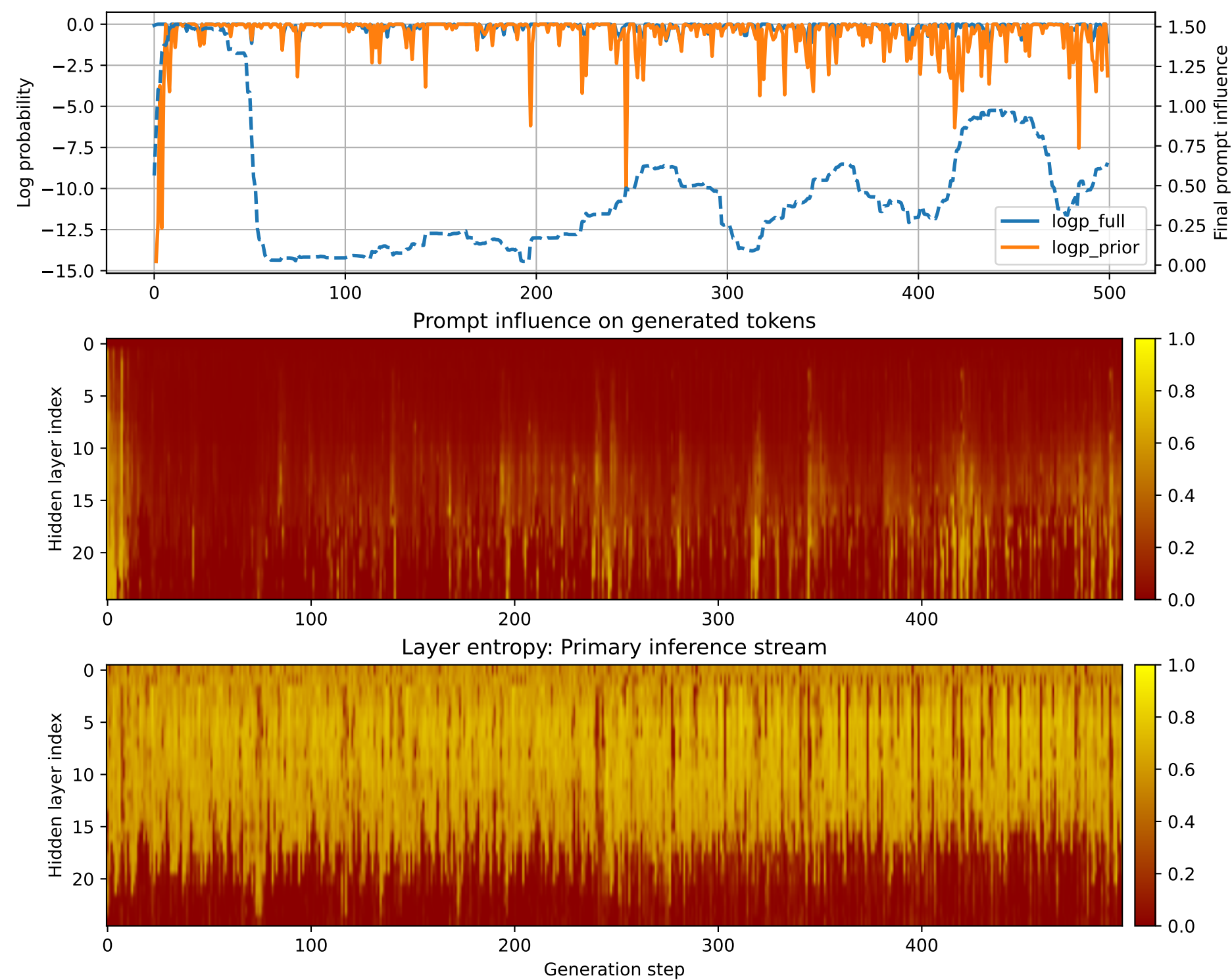


Prompt 1



PROMPT:
Task ID: PYC076 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC076 Your code must contain this required symbol:
class LRUCache Implement this task: define
class LRUCache with get and put.. Start your answer with the
check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: LRUCache ```python class LRUCache: def
__init__(self, capacity: int) -> None:
self.capacity = capacity self.cache = {} def
get(self, key: str) -> int: if key in
self.cache: return self.cache[key]
self.cache[key] = self._get_or_create(key)
return self.cache[key] def put(self, key: str, value: int)
-> None: if key in self.cache:
self.cache[key] = self._get_or_create(key) else:
self.cache[key] = value
self._put(key, value) def _get_or_create(self, key: str) ->
int: if key in self.cache:
return self.cache[key] value = self._get(key) if
value is None: self.cache[key] =
value return value def _put(self, key: str, value:
int) -> None: if key in self.cache:
self.cache[key] = self.cache.pop(key) else:
self.cache[key] = value ``` Explanation: -
The `LRUCache` class is defined with the `\_\_init\_\_` method, which
initializes an empty dictionary `self.cache`
to store the cache entries. - The `get` method takes a key as
input and returns the value associated with that
key if it exists in the cache. If the key is not found in the
cache, it calls the `\_get\_or\_create` method to
either retrieve the value from the cache or create a new entry
with the given key and its initial value. - The
`\_get\_or\_create` method checks if the key is already present in
the cache. If it is, it returns the existing
value. If the key is not present, it calls the `\_get` method to
retrieve the value from the cache and returns
it. If the value is not found, it creates a new entry with the
given key and its initial value. - The `put`
method takes a key and a value as input and updates the
corresponding value in the cache. If the key is
already present in the cache, it updates the existing value. If
the key is not present, it creates a new entry
with the given key and its initial value, and then calls the
`\_put` method to update the cache. - The `\_put`
method first